HSF-India Trainee Project Proposal

Trainee Information

- **Name:** Dhruval Patel
- **Home Institution:** Adani University
- **Project Duration:** July 13, 2026 – December 31, 2026 *(Extended through December 2026)*
- **Supervisors:**
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Project Title

Exploring Diffusion Models for Hadronic Physics Simulation

Project Description and Goals

This project aims to investigate the use of generative diffusion models for simulating hadronic interactions in high-energy particle collisions. Accurate modeling of hadronic showers is one of the main computational bottlenecks in detector simulation frameworks such as Geant4, due to the complexity and stochastic nature of the underlying physics. Recent studies have demonstrated the potential of generative AI techniques, in particular conditional normalizing flows, to learn and reproduce key features of hadronic interactions from Geant4-simulated data. While these methods show encouraging performance, certain regions of phase space, particularly at low energies, remain challenging to model accurately.

The primary goal of this project is to explore the potential of diffusion-based generative models for modeling hadronic shower development, and to investigate whether they can complement existing generative approaches, particularly in challenging regions of phase space such as the low-p_T regime. In particular, the project will focus on training conditional diffusion models on simulated hadronic interaction data and evaluating their ability to reproduce key physics observables and distributions. If successful, this approach could provide a foundation for GPU-native surrogate models of hadronic physics, enabling faster and more scalable simulation workflows for the LHC and future experiments.

Project Extension Overview (October – December 2026): Building upon successful completion of the foundational milestones—establishing the baseline diffusion architecture, custom training loops, and preliminary validation against Geant4—the project has been extended through December 2026. The extended work plan tackles critical advanced challenges: transitioning from static conditioning to continuous dynamic kinematics, enforcing strict four-vector conservation, optimizing ultra-fast inference via few-step ODE solvers, and preparing a flagship publication.

Proposed Schedule & Milestones

The project development is structured across the initial twelve-week foundational phases and the subsequent three-month extension focusing on dynamic kinematics, physics validation, and fast inference:

Phase / Period	Timeline	Key Objectives & Milestones
Phase 1	Weeks 1 – 2	Set up the working environment, secure access to initial Geant4 baseline datasets, and perform preliminary data exploration and preprocessing.
Phase 2	Weeks 3 – 4	Design the initial conditional diffusion model architecture specifically targeted at hadronic shower generation and establish baseline structure.
Phase 3	Weeks 5 – 6	Develop custom training loops, implement initial model training on high-energy physics datasets, and focus on loss optimization.
Phase 4	Weeks 7 – 8	Evaluate preliminary generation results and fine-tune hyperparameters to better model challenging phase space regions (e.g., the low-p_T regime).
Phase 5	Weeks 9 – 10	Conduct rigorous physics validation by benchmarking the diffusion surrogate model's output against standard Geant4 observables and distributions.
Phase 6	Weeks 11 – 12	Finalize code documentation, consolidate findings into a final project report, and outline abstracts for prospective external submissions.
Extension Month 1	Weeks 13 – 16 (October – Nov)	**Architectural Adaptation to Dynamic Kinematics:** Dynamic kinematic embedding (E_{inc}, η_{inc}, ϕ_{inc}) via Fourier/Sinusoidal projections and AdaLN across transformer blocks; zero-RAM streaming dataloader integration.
Extension Month 2	Weeks 17 – 20 (Nov – Dec)	**Training & Physics Validation Across Kinematics:** Train PointCloudDiT across continuous spectrum; test unseen energy interpolation (25, 50, 75 GeV); benchmark four-vector conservation balance, angular profiles, linearity, and resolution.
Extension Month 3	Weeks 21 – 24 (December)	**Fast Inference & Publication:** Benchmark few-step ODE solvers (10–15 steps) targeting ~10,000× Geant4 speedup with four-vector conservation; draft flagship publication on scaling laws, four-vector fidelity, and dynamic kinematics.

Note: The extended milestones for Months 1–3 represent the continued development plan agreed upon with project mentors to expand model capabilities from fixed single-energy showers to dynamic, full-detector kinematics. Full details for each phase are detailed in the following section.

Extended Work Plan (October 2026 – December 2026)

During this extended three-month period, the project focuses on advancing the PointCloudDiT generative framework to continuous dynamic kinematics, verifying rigorous physical conservation laws, achieving orders-of-magnitude faster generation, and compiling findings for peer-reviewed publication.

Month 1: Architectural Adaptation to Dynamic Kinematics

1. Continuous Kinematic Conditioning:

- Replace fixed conditioning with a dynamic kinematic embedding vector (E_{inc}, η_{inc}, ϕ_{inc}).
- Implement continuous Fourier / Sinusoidal projections coupled with Adaptive Layer Normalization (AdaLN) across the transformer blocks to dynamically modulate latent representations.

2. Dataset & Dataloader:

- Integrate multi-energy calorimeter events into our zero-RAM streaming dataloader across the designated kinematic range.

Month 2: Training & Physics Validation Across Kinematics

1. Model Training:

- Train PointCloudDiT across the continuous kinematic spectrum using our optimized scaling configuration.

2. Interpolation Tests:

- Test the model on unseen intermediate energy values (e.g., evaluate at 25 GeV, 50 GeV, and 75 GeV) to verify smooth physical generalization without mode collapse.

3. Physics & Conservation Benchmarks:

- **Angular Distributions (η and ϕ):** Evaluate transverse shower profiles along pseudorapidity (η) and azimuthal angle (ϕ) to verify that directional shower development and beam-tilt angles are accurately modeled.
- **Four-Vector Conservation ($p^{\mu} = (E, p)$):** Rigorously verify energy-momentum balance by reconstructing total four-momentum from generated hits ($\sum p_{hits}^{\mu}$) against the incident particle four-vector ($p_{incident}^{\mu}$).
- **Energy Linearity & Resolution:** Confirm response linearity $\langle E_{visible} \rangle \propto E_{inc}$ and fit standard calorimeter resolution parameters $\sigma(E)/E = a/\sqrt{E} \oplus b$.
- **Longitudinal Depth:** Confirm the logarithmic progression of shower maximum depth ($Z_{max} \propto \ln E_{inc}$).

Month 3: Fast Inference & Publication

1. Fast Generation (DPM-Solver):

- Benchmark few-step ODE solvers (10–15 steps) across the dynamic range to achieve ~10,000× speedup over Geant4 while preserving four-vector conservation.

2. Paper Preparation:

- Draft a flagship publication consolidating our 3×3 scaling laws, four-vector fidelity, and dynamic kinematic results for submission to a top journal or conference.